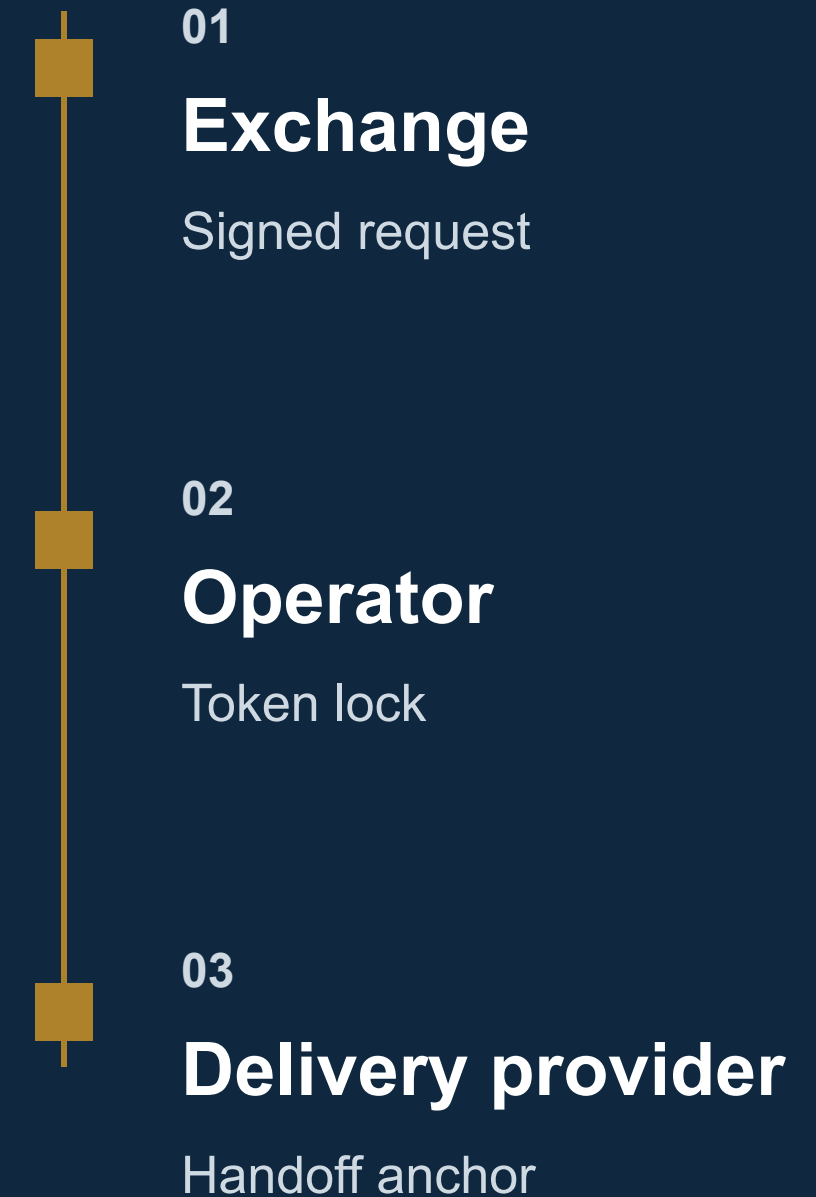


Make every redemption delay attributable.

Independent evidence for the journey
from a token lock to a delivery claim.

A commitment to record and prove each step.
Redemption approval remains an institutional decision.

ESCROW / EXCHANGE INBOX / PUBLIC VERIFIER



One redemption journey. Three accountable clocks.

01

Locate the delay

Separate exchange forwarding, operator decision-making, and delivery-provider recording into distinct obligations.

02

Keep evidence portable

Let holders check signed records and public commitments without relying on the institution's server.

03

Test before adoption

Use reproducible local scenarios and verified testnet deployments as the starting point for a partner pilot.

Implemented today: signed receipts + public log + redemption escrow + exchange inbox.

A token lock reveals custody— not who is delaying redemption.

Holders see one delay. Responsibility spans several institutions.



VISIBLE ONCHAIN

Lock, amounts, token movements

EVIDENCE GAP

Decision timing, handoff, delivery claims

Start with holders who can sign and institutions that can integrate.

PRIMARY USER

Self-custody holders and institutions

Direct locking starts the operator's clock.
The holder retains control of challenges,
reclaims, disputes, and acknowledgement.

ADOPTER

Issuer / operator

Integration incentive: distinguish responsibility
across the redemption chain. Hypothesis, not traction.

CONDITIONAL

Exchange-wallet customer

Needs an external signing key. The inbox does not
verify customer status, balance, or eligibility.

OUTSIDE DIRECT SCOPE

Custodial user without a key

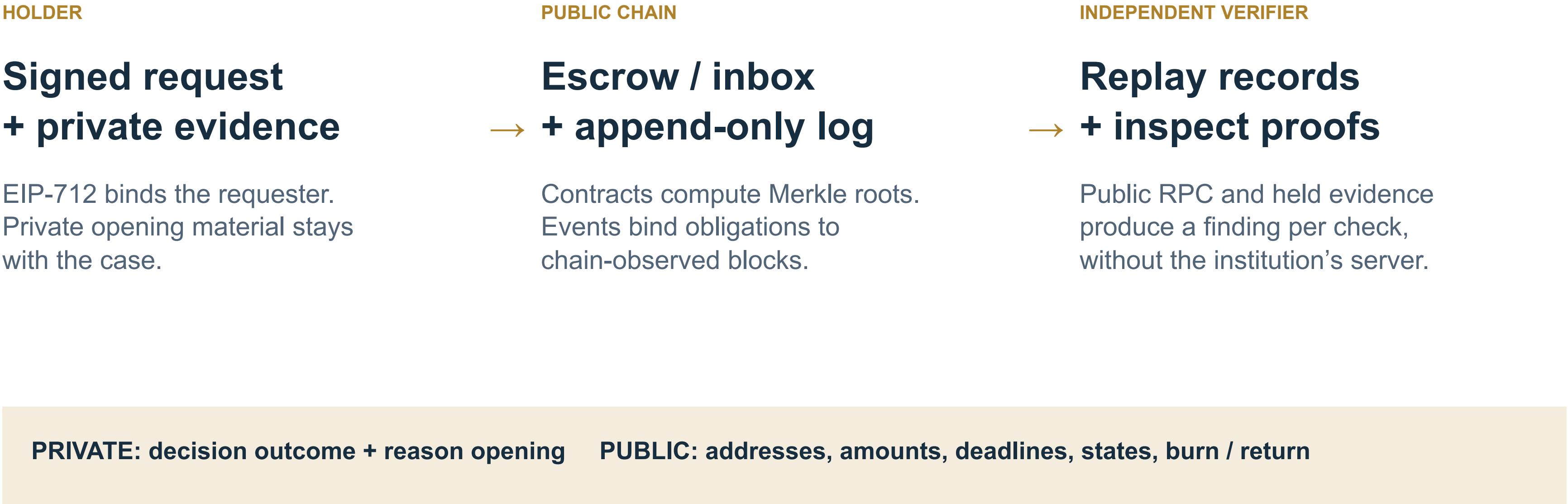
Cannot independently sign the current flow.
Passkeys and gas sponsorship are not implemented.

Each clock starts from a different onchain event.

RESPONSIBLE PARTY	CLOCK STARTS AT	REQUIRED EVIDENCE	IF UNANSWERED
01 Exchange	Holder-signed request submitted to inbox	Forward into escrow or prove rejection	Exchange breach only; operator clock not started
02 Operator	Token lock (direct or via inbox)	Decision record; then a proven handoff	finalize returns tokens after the response window
03 Delivery provider	Handoff anchor recorded onchain	Delivery claim in provider's own log	STALLED; tokens remain locked

Requests and locks start clocks without waiting for a separate acceptance receipt.

Public commitments make records independently checkable.



Return, hold, or burn— each follows explicit conditions.

RETURN

**Operator non-response is finalized,
or the handoff deadline passes.**

Tokens return through finalize / reclaim.

HOLD

**Provider non-response → STALLED.
Holder dispute → DISPUTED.**

Tokens stay locked pending offchain
resolution.

BURN

**Provider delivery claim is proven;
holder acknowledges or dispute window expires.**

After expiry in DELIVERED, anyone can
call burn.

The dispute window begins when the escrow accepts the proof—not at an older delivery anchor.

Missing evidence must remain distinct from a proven breach.

CONFIRMED

Signature / binding / proof checks out

NOT_DUE

Deadline has not arrived

OBLIGATION_UNMET

Inspect the failed check and evidence

UNVERIFIABLE

Evidence is insufficient

OUT_OF_SCOPE

Fact is outside the tool’s guarantees

THREATS ADDRESSED

Institution: omission and lateness
Requester: fabricated obligations
Third party: forged bundles

OUTSIDE THE GUARANTEE

Physical gold and actual delivery
Truth of private reasons
Legal enforcement and chain finality

The prototype is reproducible; production validation is still ahead.

191

Automated tests

107 Solidity + 84 TypeScript

30

Local scenarios

20 attack + 10 honest

3

Source-verified contracts

Sourcify exact_match on Sepolia

LOCAL EVIDENCE

All 30 scenarios matched expected findings.
Zero expectation mismatches in honest cases.

This is not a statistical false-positive rate.

PUBLIC EVIDENCE

Six deployment / registration transactions.
MockGold + escrow + inbox on Sepolia.

No physical redemption or full public scenario run.

The contribution is accountability for offchain decisions.

REFERENCE	CORE MECHANISM	BOUNDARY / ADAPTATION
Rollup inboxes	Forced inclusion	The rollup protocol can enforce inclusion. An offchain service cannot be forced the same way.
Transparency logs	Append-only commitments and inclusion proofs	Users monitor their own entries; public roots provide evidence of inclusion.
BlockNotice	Bound receipts, chain-derived deadlines, asset locks	Locks start the operator clock without a receipt; handoff starts the provider clock.

Adapt established primitives to make delay attribution independently verifiable.

Adoption depends on identity, disputes, and operational controls.

01

Identity & access

Authenticate institutions and customers.
Address inbox spam and false balance claims.
Plan key rotation and wallet support.

02

Disputes & incentives

Define offchain resolution for locked cases.
Validate issuer and legal-team incentives.
Evaluate institutional collateral / penalties.

03

Operational readiness

Set finality and monitoring policies.
Audit contracts and token assumptions.
Pilot with a real operating partner.

Shipped: working prototype and testnet evidence. Next: validate the operating model with a partner.

Verify the evidence. Then test the operating model.

REPRODUCE THE LOCAL DEMO

npm run demo

CHECK THE PUBLIC DEPLOYMENT

npm run check:escrow -- sepolia

Setup and SEPOLIA_RPC_URL are documented in the README.

SEPOLIA / 11155111

MockGold	0xb3a791...955b1a
Escrow	0x8c8cbf...f7778c
Inbox	0x870238...bb5d

Three services, one demo wallet.
MockGold has no physical-gold backing.
Source verification is not a security audit.

github.com/tnwjd023-boop/BlockNotice_GoldRWA